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SECTION : BSE – 4A

Opearting Systems Lab 13

BSSE SPRING 2026

Lab 13

Question 01:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

sem_t sem1, sem2;

void *thread1(void *arg)
{
    for (int i = 0; i < 5; i++)
    {
        sem_wait(&sem1);
        printf("Thread 1 Running\n");
        sleep(1);
        sem_post(&sem2);
    }
    return NULL;
}

void *thread2(void *arg)
{
    for (int i = 0; i < 5; i++)
    {
        sem_wait(&sem2);
        printf("Thread 2 Running\n");
        sleep(1);
        sem_post(&sem1);
    }
    return NULL;
}

int main()
{
    pthread_t t1, t2;

    sem_init(&sem1, 0, 1);
    sem_init(&sem2, 0, 0);

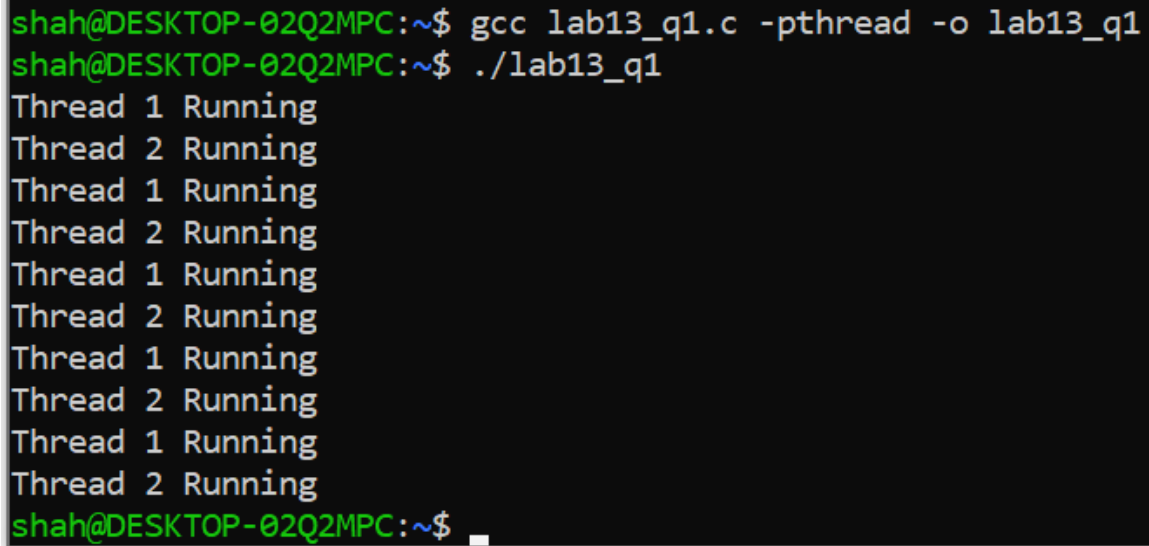
    pthread_create(&t1, NULL, thread1, NULL);
```

```
pthread_create(&t2, NULL, thread2, NULL);

pthread_join(t1, NULL);
pthread_join(t2, NULL);

sem_destroy(&sem1);
sem_destroy(&sem2);

return 0;
}
```



```
shah@DESKTOP-02Q2MPC:~$ gcc lab13_q1.c -pthread -o lab13_q1
shah@DESKTOP-02Q2MPC:~$ ./lab13_q1
Thread 1 Running
Thread 2 Running
Thread 1 Running
Thread 2 Running
Thread 1 Running
Thread 2 Running
Thread 1 Running
Thread 2 Running
Thread 1 Running
Thread 2 Running
shah@DESKTOP-02Q2MPC:~$
```

Question 02:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <stdlib.h>
#include <unistd.h>
```

```
sem_t empty, full;
int item;
```

```
void *producer(void *arg)
{
    for (int i = 0; i < 5; i++)
    {
        sem_wait(&empty);
```

```
    item = rand() % 100;
    printf("Produced: %d\n", item);

    sleep(1);

    sem_post(&full);
}
return NULL;
}

void *consumer(void *arg)
{
    for (int i = 0; i < 5; i++)
    {
        sem_wait(&full);

        printf("Consumed: %d\n", item);

        sleep(1);

        sem_post(&empty);
    }
    return NULL;
}

int main()
{
    pthread_t p, c;

    sem_init(&empty, 0, 1);
    sem_init(&full, 0, 0);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);

    pthread_join(p, NULL);
    pthread_join(c, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
}
```

```
return 0;
}
shah@DESKTOP-02Q2MPC:~$ gcc lab13_q2.c -pthread -o lab13_q2
shah@DESKTOP-02Q2MPC:~$ ./lab13_q2
Produced: 83
Consumed: 83
Produced: 86
Consumed: 86
Produced: 77
Consumed: 77
Produced: 15
Consumed: 15
Produced: 93
Consumed: 93
shah@DESKTOP-02Q2MPC:~$
```

Question 03

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
```

```
sem_t mutex;
int totalSum = 0;
```

```
int arr[] = {1,2,3,4,5,6,7,8,9,10};
```

```
struct Data
{
    int start;
    int end;
};
```

```
void *sumArray(void *arg)
{
    struct Data *d = (struct Data *)arg;
    int localSum = 0;

    for (int i = d->start; i < d->end; i++)
        localSum += arr[i];
}
```

```
sem_wait(&mutex);
totalSum += localSum;
sem_post(&mutex);

return NULL;
}

int main()
{
    pthread_t t1, t2;
    struct Data d1 = {0,5};
    struct Data d2 = {5,10};

    sem_init(&mutex, 0, 1);

    pthread_create(&t1, NULL, sumArray, &d1);
    pthread_create(&t2, NULL, sumArray, &d2);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Total Sum = %d\n", totalSum);

    sem_destroy(&mutex);

    return 0;
}
```

```
shah@DESKTOP-02Q2MPC:~$ gcc lab13_q3.c -pthread -o lab13_q3
shah@DESKTOP-02Q2MPC:~$ ./lab13_q3
Total Sum = 55
shah@DESKTOP-02Q2MPC:~$
```

Question 04

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <limits.h>

sem_t mutex;
int minimum = INT_MAX;
```

```
int arr[] = {25,10,45,3,17,8,99,1,50,6};
```

```
struct Data  
{  
    int start;  
    int end;  
};
```

```
void *findMin(void *arg)  
{  
    struct Data *d = (struct Data *)arg;  
    int localMin = INT_MAX;  
  
    for (int i = d->start; i < d->end; i++)  
    {  
        if (arr[i] < localMin)  
            localMin = arr[i];  
    }  
  
    sem_wait(&mutex);  
  
    if (localMin < minimum)  
        minimum = localMin;  
  
    sem_post(&mutex);  
  
    return NULL;  
}
```

```
int main()  
{  
    pthread_t t1, t2;  
    struct Data d1 = {0,5};  
    struct Data d2 = {5,10};  
  
    sem_init(&mutex, 0, 1);  
  
    pthread_create(&t1, NULL, findMin, &d1);  
    pthread_create(&t2, NULL, findMin, &d2);  
  
    pthread_join(t1, NULL);
```

```
pthread_join(t2, NULL);

printf("Minimum Value = %d\n", minimum);

sem_destroy(&mutex);

return 0;
}
```

```
shah@DESKTOP-02Q2MPC:~$ gcc lab13_q4.c -pthread -o lab13_q4
shah@DESKTOP-02Q2MPC:~$ ./lab13_q4
Minimum Value = 1
shah@DESKTOP-02Q2MPC:~$
```